

HILLSBORO, TX, USA

WMO: 720295

Lat: 32.084N

Lon: 97.097W

Elev: 209

StdP: 98.84

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53972

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) -2.1	(d) -11.3	(e) 1.5	(f) 0.7	(g) -9.0	(h) 1.8	(i) 2.1	(j) 12.8	(k) 13.8	(l) 11.7	(m) 12.0	(n) 3.2	(o) 350	(p) 0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.7	(c) 37.8	(d) 22.7	(e) 37.0	(f) 22.9	(g) 35.9	(h) 23.0	(i) 25.6	(j) 31.5	(k) 25.2	(l) 31.3	(m) 24.7	(n) 30.9	(o) 4.1	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.6	27.1	23.6	18.9	27.1	23.0	18.2	27.1	79.6	31.3	77.7	31.4	75.9	30.9	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.1	(b) 9.8	(c) 8.7	(e) -8.6	(f) 39.4	(g) 2.0	(h) 1.2	(i) -10.0	(j) 40.3	(k) -11.1	(l) 41.0	(m) -12.2	(n) 41.6	(o) -13.7	(p) 42.5		
(a) 11.1	(b) 9.8	(c) 8.7	(e) -8.7	(f) 25.8	(g) 2.2	(h) 0.8	(i) -10.3	(j) 26.5	(k) -11.6	(l) 26.9	(m) -12.8	(n) 27.4	(o) -14.4	(p) 28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	8.4	10.6	14.9	18.7	22.8	27.5	28.9	29.4	26.4	20.1	14.3	9.4
	DBStd	8.61	5.41	6.01	5.17	4.29	3.58	2.26	1.97	2.33	3.11	4.82	5.38	5.35
	HDD10.0	278	96	61	18	2	0	0	0	0	0	2	21	78
	HDD18.3	1176	309	222	128	46	8	0	0	0	0	38	141	282
	CDD10.0	3680	46	78	169	264	398	525	586	602	491	314	149	58
	CDD18.3	1537	2	6	21	59	148	275	328	343	241	92	19	4
	CDH23.3	17237	17	36	121	419	1277	3161	4109	4477	2632	851	117	22
CDH26.7	8486	1	2	14	97	442	1569	2215	2531	1304	289	20	2	
Wind	WSAvg	4.0	4.0	4.5	4.9	4.9	4.2	3.9	3.5	3.4	3.3	3.8	4.1	3.9
Precipitation	PrecAvg	1000	74	76	98	87	101	89	48	65	85	110	81	82
	PrecMax	1388	136	188	270	185	236	273	118	158	270	321	233	240
	PrecMin	600	13	3	14	31	21	22	0	1	9	3	15	10
	PrecStd	218	36	46	57	48	55	69	33	46	62	79	60	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	27.0	28.3	31.4	34.8	37.8	38.8	40.0	37.8	33.9	29.3	26.1
		MCWB	15.3	16.5	18.0	20.4	22.9	22.8	22.9	22.3	21.6	21.6	20.0	17.6
	2%	DB	22.6	24.2	26.3	29.2	32.4	36.3	37.5	38.0	36.1	31.6	26.4	22.7
		MCWB	14.7	16.9	18.3	19.7	22.3	22.9	22.6	22.9	22.3	21.9	18.4	16.9
	5%	DB	20.1	22.2	24.8	27.4	31.0	35.0	36.3	37.2	34.8	30.0	24.2	20.5
		MCWB	14.7	16.0	17.8	19.6	22.2	22.9	22.9	23.1	22.5	20.6	18.0	15.8
	10%	DB	17.7	20.1	22.6	25.9	29.2	33.1	35.0	36.0	32.9	27.9	22.5	18.1
		MCWB	13.3	15.7	16.8	19.1	21.8	23.2	23.3	23.3	22.0	19.3	17.5	14.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	21.0	21.1	23.4	25.3	25.8	26.3	25.9	25.3	24.6	22.4	20.3
		MCDB	21.5	23.0	24.7	27.4	29.3	32.3	31.1	32.5	30.4	28.4	25.9	22.9
	2%	WB	17.6	19.3	20.2	22.1	24.2	25.2	25.6	25.3	24.8	23.8	20.9	18.8
		MCDB	20.1	21.8	23.8	26.7	28.6	31.5	31.2	32.4	30.1	27.8	23.3	21.6
	5%	WB	16.0	17.8	19.3	21.1	23.5	24.7	25.2	24.9	24.2	22.9	19.8	17.0
		MCDB	18.9	20.7	22.9	25.4	28.1	30.8	31.2	32.1	29.7	27.4	22.9	19.4
	10%	WB	13.6	16.2	18.2	20.3	22.7	24.1	24.6	24.5	23.6	21.9	18.5	15.2
		MCDB	17.6	19.4	21.6	24.4	27.5	30.0	30.7	31.6	29.3	26.3	21.8	17.9
Mean Daily Temperature Range	5% DB	MDBR	11.7	11.1	11.4	11.4	10.7	10.9	11.3	11.7	11.6	12.1	11.7	11.2
		MCDBR	15.2	13.3	12.0	12.0	12.1	12.8	12.9	13.2	13.2	12.7	12.7	13.8
		MCWBR	9.2	8.1	6.5	6.0	4.4	3.3	3.1	3.2	3.8	5.1	6.9	8.9
	5% WB	MCDBR	12.1	11.0	10.1	9.9	9.9	10.1	10.5	11.3	10.5	10.4	10.0	11.4
		MCWBR	9.1	7.9	6.6	5.8	4.5	3.3	3.3	3.0	3.4	4.9	7.3	9.2
Clear-Sky Solar Irradiance	taub	0.314	0.332	0.354	0.396	0.423	0.432	0.439	0.436	0.399	0.361	0.341	0.320	
	taud	2.506	2.459	2.416	2.323	2.304	2.313	2.332	2.320	2.373	2.465	2.467	2.491	
	Ebn at Noon	912	925	926	895	868	856	848	847	868	881	871	883	
	Edh at Noon	86	100	112	128	132	130	127	127	116	98	89	84	
All-Sky Solar Radiation	RadAvg	2.93	3.40	4.48	5.43	5.83	6.57	6.71	6.18	5.20	4.12	3.09	2.58	
	RadStd	0.29	0.56	0.37	0.44	0.52	0.58	0.47	0.39	0.43	0.61	0.39	0.34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page